

Toward Best Practices of Implementing Campus Sustainability in US Universities

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Abstract

As leaders in research and innovation, universities increasingly have a responsibility to recognize their environmental impacts and to advance sustainable campuses in order to embody environmental stewardship and serve as examples of

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how to integrate sustainable practices into our infrastructure and daily lives. The move toward creating green campuses has grown within the last 15 years and universities are developing new environmental offices and crafting their own sustainability plans. Our research has gathered information from universities in the United States that have demonstrated sustainability efforts and have incorporated an environmental focus into their academic programs, engagement practices, operations, and leadership. This information includes office structures, academic programs and research, student and faculty engagement, community outreach, and innovative programming. Using case studies, this chapter highlights some best practices across US universities and points to some trends in sustainability campus organization and leadership.

Keywords

Campus sustainability · HEI · Framework for organizing sustainability office structures · Leadership for sustainability · US universities · Case studies · Indices for campus sustainability · STARS · AASHE

1 Introduction: The Changing Nature of Campus Sustainability

Sustainable campus operations play a central role at the interface of climate change and education for sustainable development. Much has been written about the role of higher education in sustainable development and its contribution to achieving the Sustainable Development Goals (SDGs), as set by the United Nations' *2030 Agenda for Sustainable Development* (Ferrer-Estévez and Chalmeta 2021; Hernandez et al. 2020; Leal Filho 2020; Leal Filho et al. 2021; United Nations 2015; Weiss et al. 2021). Papers acknowledge the advantages of embedding the SDGs in key campus areas (Karatzoglou 2013; Leal Filho et al. 2019a). An extensive review of the literature shows that there is a huge body of academic research on how campus sustainability has been evolving as a topic in both research and practice, how it has been reflected into university plans and the SDGs to date, which challenges hinder their implementation, and how to overcome those challenges (Leal Filho et al. 2017, 2018, 2021).

It is important to review campus sustainability as universities have often been described as representations of “small cities” with all the issues and opportunities that exist in them (Alshuwaikhat and Abubakar 2008). They have also been designated as “living laboratories” – places that promote exploration and creativity and leave a “revolving door” between research, education, and campus systems (Leal Filho et al. 2015). Moreover, universities that have successfully implemented sustainability on their campuses serve as “learning institutions” – those that have undertaken the responsibility to train the future generations about the importance and applications of sustainability (Barlow and Stone 2011). It has been stated that universities have both an instrumental and functional role to play in the SDGs as

education should be a goal in itself, as described by SDG4, but it is also a guiding tool for all the other SDGs (Ferrer-Estévez and Chalmeta 2021).

Campus sustainability has been defined as efforts on the part of Higher Education Institutions (HEI) to manage environmental issues and promote social change (Disterheft et al. 2013; Washington-Ottombre et al. 2018). Campus sustainability is evolving as a concept; it is being implemented in different ways, and the duties and responsibilities of staff who are engaged in sustainability efforts at universities have been changing. Campus sustainability has transitioned from one-off initiatives focused on “educating agents of change and modeling change” to implementing change more holistically; likewise, the types of offices and the leadership structures that are in charge of setting the goals and the mission of universities as sustainable organizations have evolved in recent years (Washington-Ottombre et al. 2018). In fact, universities do not share a one-size-fits-all organizational structure. Some have facilities and operations offices that are separate from academic programs and research, others have a sustainability governance structure that is represented by committees, and still others have centralized operations, where both academics and facilities are directed by and report to a single office or leader.

Focusing on a select number of universities, this chapter assesses the various types of administrative structures and organizations that are in charge of implementing sustainability at US universities and considers their relationship to the success of sustainability performance. In particular, it analyzes the nature of university sustainability offices and the implications of sustainability leadership on reaching sustainability targets and implementing campus sustainability goals. To this end, this chapter:

1. Reviews the elements of sustainable campus indicators and the models for assessing effectiveness
2. Describes existing organizational structures and functions as they relate to sustainability leadership
3. Highlights some best practices across US universities, pointing to trends in sustainability campus organization and leadership via case studies

This chapter is organized into eight sections. Section 2 examines the dimensions of campus sustainability. Sections 3 and 4 explore the role of leadership in campus sustainability and present a framework for organizing the various leadership structures. Section 5 focuses on the methodology and the use of an indicator-based approach for assessing sustainability implementation effectiveness. Section 6 presents the case studies of select US universities. Sections 7 and 8 discuss the findings and present the conclusions.

2 The Variable Dimensions of Campus Sustainability

No single paper states how campus sustainability is or should be organized institutionally. Campus sustainability has been defined as centering on three main dimensions: physical, educational, and institutional (Disterheft et al. 2013; Washington-Ottombre et al. 2018). Starting from the premise that universities should operate as fully integrated communities modeled after social and biological sustainability, Cortese (2003) proposes a university system model with four interdependent dimensions, namely Education, University Operations, Research, and External Community. Since these dimensions must also be monitored, assessed, and reported on continuously, Lozano (2006b) and Lozano et al. (2013) add a fifth dimension – Assessment and Reporting. Aiming to identify the different practices and domains of sustainability among HEIs, Lozano et al. (2015) later on use seven dimensions – Institutional framework; Campus operations; Education; Research; Outreach and collaboration; On-campus experiences; and Assessment and reporting.

In a comprehensive framework that integrates the SDGs in universities, Leal Filho et al. (2021) propose four key areas of sustainability organization: Research and development; Learning and teaching; Governance and campus operations; and Civic engagement and community outreach. Velazquez et al. (2005, 2006) point out to similar dimensions as strategies for fostering sustainability in HEIs. Other authors, such as Gomez et al. (2015), also propose four dimensions, but with differently organized and named categories – Education & Research, Public Engagement, Operations, and Administration. When presenting their four dimensions in a schematic model, the authors place the Administration dimension in the center and explain that it covers all policies, strategies, and high-level decisions influencing all other dimensions. They recognize that while uncommon in other sustainability frameworks, the Administration dimension should be recognized as an important component in the implementation of sustainability initiatives. Thus, over the last two decades, the literature on campus sustainability has: (1) articulated a variety of dimensions; (2) described its complexity in increasing detail; and (3) focused on the role of decision-making and integration of the various dimensions.

3 The Transformational Role of Leadership and Planning for Campus Sustainability

The role of leadership, planning, and administration has been reviewed in many papers and assessment frameworks. Investigating the challenges in achieving sustainability in higher education, Leal Filho et al. (2020) review the differences between management and leadership and conclude that “Higher Education Institutions are increasingly taking leadership on sustainable development more seriously and reflecting on them” (p. 12). Universities are recognized for institutionalizing sustainability by dedicating resources to sustainability coordination, developing plans to move toward sustainability, and engaging stakeholders in governance

(AASHE 2019c). In short, three main elements are used for assessing leadership in campus sustainability: planning, coordination, and governance.

It has been acknowledged that good coordination efforts are needed in order to organize, implement, and spread the word about sustainability initiatives across campus and beyond. Good coordination necessitates dedicated position(s) that are specifically tasked with carrying out the sustainability efforts. In most US universities, such positions are part of the sustainability office, whose primary task is to determine guidelines and set measurable standards related to operations, planning, and execution of projects that advance campus sustainability. The office of sustainability is said to serve as “a node, from where all sustainability related activities are coordinated” (Leal Filho et al. 2019a, p. 1396). In some cases, as a part of operations, sustainability employees have designated functions related primarily to facility operations, i.e., reducing the university carbon footprint and switching to renewable energy sources, increasing energy efficiency, reducing water consumption, providing sustainable food choices, ensuring availability of green spaces on campus, etc. Such efforts typically lead to measurable sustainability-related changes, when carried out across the campus and when they permeate all of the campus offices and tasks. In cases where only a particular department or school has a sustainability officer whose efforts are confined to that particular establishment, the university as a whole has a difficult time advancing overall campus sustainability. Therefore, employees or committees tasked with the coordination of sustainability initiatives have to be given clear mandates by the executive leadership or board of trustees. In some universities, it is the sustainability office staff who participate in committees and working groups that have a policy-making function and advise leadership on sustainability implementation.

Sustainability planning plays a significant role in assessing campus sustainability performance as it “affords an institution the opportunity to clarify its vision of a sustainable future and provides a roadmap to help guide decision-making” (AASHE 2019c). In addition, sustainability planning sets an institution’s vision of a sustainable future, establishes priorities, and helps guide budgeting and decision-making (Leal Filho et al. 2019a). Universities that include sustainability in their strategic plan and other guiding documents signal their dedication and concerted efforts to advance sustainability, which may help “infuse an ethic of environmental, fiscal and social responsibility throughout the campus community” (AASHE 2019c).

One measure of successful campus sustainability implementation is the number and type of sustainability statements and commitments. To show commitment for developing sustainability, many universities have signed pledges, sustainability plans, and statements. A review paper maintains that universities that have signed such declarations, plans, and initiatives are more likely to implement sustainability on their campuses (Lozano et al. 2015). Although it is unclear whether signing a pledge leads to commitment or vice versa, the authors maintain that expressing commitment is a step in the right direction toward making sustainability a central issue that requires attention at all levels. Another way for signaling commitment to both internal and external stakeholders is by developing and implementing a

sustainability plan or climate action plan, or by including sustainability as part of the overall strategic plan of the institution.

Stakeholder engagement is vital in advancing sustainability. Numerous papers focus on the impact a particular group of stakeholders has on campus sustainability. Some analyze student perceptions and involvement in creating sustainability tools and practices (Abubakar et al. 2016; Adams et al. 2011; Barth and Timm 2010). Others talk about the role of alumni, and still others discuss the role of all stakeholders (Brinkhurst et al. 2011). Abubakar et al. (2016) maintain that achieving campus sustainability more effectively is not possible without the cooperation and involvement of all stakeholders such as students, faculty and staff, university management, funding agencies, and the community in the ongoing governance of HEI. Moreover, the participation of all stakeholders contributes to creating a culture of engagement (Washington-Ottombre et al. 2018).

Finally, governance includes a variety of organizational functions and decision-making processes, from financial oversight and personnel management to goal-setting and strategic planning. Inclusive and participatory processes and structures that empower stakeholder groups to work collaboratively in order to address sustainability challenges are crucial to the success of sustainability implementation. As stated by AASHE, “without transformed governance structures, many sustainability gains cannot be realized” (2020, p. PA-03-1). Moreover, for universities to be able to truly implement sustainability, they need to employ transformative leaders who can “build a shared vision with their followers, who in effect become leaders and contribute to the process” (Mader 2012, p. 83).

4 Operationalizing Leadership for Campus Sustainability

As discussed earlier in this chapter, there is a paucity of research on the various types of responsibilities of those who are tasked with implementing campus sustainability. Although the literature provides a glimpse into the role of institutional leaders for promoting campus sustainability (Leal Filho et al. 2020), it does not answer questions about the relationship between sustainability leadership and organizational effectiveness in HEI. For example, it does not investigate whether sustainability offices that separate operations and facilities function from academic programs and research are more successful in implementing campus sustainability overall than universities that integrate the operations and academic functions, or those that rely on committees for making and implementing most sustainability-related decisions and goals. To bring greater clarity to the linkage between sustainability leadership and organizational effectiveness, we have devised the following framework for organizing different sustainability office structures.

We have categorized three types of organizational structures based on the extent to which offices are tasked with and carry out their responsibilities in the areas of planning, coordination, and governance. A summary of the three structures is presented in Fig. 1.

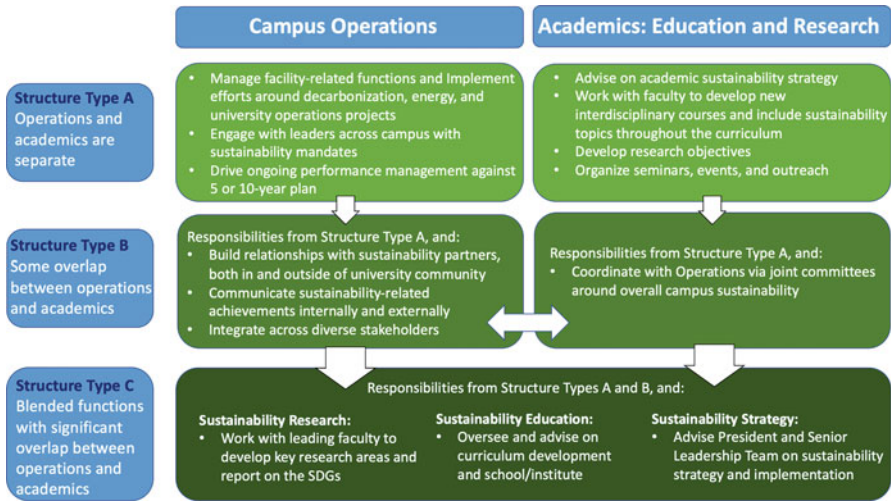


Fig. 1 Sustainability structure types and their primarily responsibilities. (Designed by Alexia Stefanovich, <https://right-eye-studios.com>)

Figure 1 shows that universities under structure Type A employ sustainability staff whose primary responsibility is to manage facilities-related functions and implement climate-related goals to buildings and natural resource efficiency use, e.g., water and energy. Under this structure, the facilities office staff is organized separately and has minimal interaction with sustainability education and research. The academic lead is engaged in education and research of sustainability, has an advisory role in creating curriculum and setting the academic sustainability strategy in terms of number and type of sustainability courses offered and hiring sustainability-related faculty and staff.

Under structure Type B, in addition to the functions executed by the typical A structure, other responsibilities are added. Campus operations staff engage with other departments to communicate about the importance of sustainability, to build relationships with sustainability partners, and to work with interns to communicate sustainability achievements to internal and external campus communities. Under structure Type B, the academic lead coordinates with facilities and operations via joint committees related to overall campus sustainability. These joint committees are established with broad campus participation from students, staff, and faculty. At least one sustainability committee, office, and/or officer is tasked by the administration or governing body to advise on and implement policies and programs related to sustainability on campus. This committee, office, and/or officer focuses on sustainability broadly (i.e., not just on one sustainability issue, such as climate change) and covers the entire institution.

In structure Type C, there is an overlap of the duties and activities of the operations and academic leads. The sustainability lead works with faculty to develop key research areas and integrate campus facilities and operations project data into

education and research. Multiple committees or offices are tasked either with the overall campus sustainability or with responsibilities for the different departments or schools; in both cases, they coordinate across the university. Committees have broader participation, with wider representation from internal and external stakeholders who have a regular voice in institutional decisions that affect them. In some instances, there is significant partnership and coordination among the different committees and their number increases. Typically, they prepare or adopt a sustainability action plan, help with curriculum development, advise senior leadership on sustainability strategy and implementation, and engage with stakeholders to build lasting relationships.

5 Methodology

5.1 Using an Indicator-Based Approach to Assess Sustainability Effectiveness

Many comprehensive models for measuring the dimensions of campus sustainability have been proposed in the literature (Gómez et al. 2015; Hernández-Díaz et al. 2021; Lozano 2006a). One of them is the indicator-based approach, which uses narratives, data, and maps – all organized by indicators that are measurable. This enables organizations to self-report data in specific categories, for which later on they are awarded points. Compared to other approaches, this has the highest potential for transparency and consistency, and the most usefulness for decision makers (Lozano 2006a).

This chapter uses the STARS indicator-based approach as a proxy of effective sustainability implementation and assumes that universities with higher ratings have implemented sustainability more efficiently on their campus. STARS, which stands for Sustainability Tracking and Assessment Rating System, was developed and implemented by the Association for the Advancement of Sustainability in Higher Education (AASHE). It is organized into and awards points within four main categories – Academics, Engagement, Operations, and Planning & Administration (AASHE 2020c). A fifth category – Innovation & Leadership – has been added to recognize new and path-breaking practices that are not covered by other STARS credits (AASHE 2019c).

As AASHE uses an open consultation process, the STARS model is continually being improved. In addition, the scores that are awarded are valid for 3 years, after which the data are updated and re-assessed. Although the model has become “one of the most comprehensive and popular tools used today” (Hernandez-Díaz et al. 2021; Saadatian et al. 2011; Gomez et.al. 2015), it has been criticized for requiring a paid membership for participation and for being centered on and most applicable to North American universities, “where sustainability efforts are considered to be more widespread and supported by the institutions themselves” (García 2010). Researchers label STARS as “useful for supporting the sustainability journey of HEIs, working as a road map where the experiences of more advanced institutions

are presented through the proposed indicators, rationale and criteria” (Gomez et al. 2015, p. 478). Although we are not aware of other research papers that use the STARS model as a proxy of sustainability implementation effectiveness, we believe it can serve as an appropriate measure, especially for research like ours that focuses on universities located in North America.

To assist institutions with identifying specific contributions, AASHE publishes the STARS Technical manual, which includes information about which STARS subcategories (e.g., Curriculum, Public Engagement, Energy, etc.) contribute most directly to assessing sustainability implementation and, as of recently, evaluating performance on the SDGs (AASHE 2019c). Therefore, universities that use the STARS reporting system can easily “translate” their performance to be commensurate with the SDGs. Reciprocally, universities that report on the SDGs typically have more sophisticated data collection methods and reporting tools and have thought about the development and implementation of sustainability-related plans.

Another advantage of the STARS model is its comprehensive scoring system and ranking of universities in several categories, called impact areas. In its Sustainable Campus Index, AASHE identifies top-performing colleges and universities for their overall performance by Institution Type in four classifications areas: Associate Colleges, Baccalaureate, Master’s, and Doctoral Institutions (AASHE 2020c). Based on data submitted by the institutions themselves, universities score points that lead to a rating of Bronze (25–44 points), Silver (45–64 points), Gold (65–84 points), or Platinum (85+), or are recognized as a STARS Reporter – a category that describes institutions that wish to participate in STARS but do not want to pursue an overall STARS rating or make their scores public. Currently, 1,030 institutions are registered to use the STARS Reporting Tool, of which 674 have earned a STARS rating (AASHE 2021).

5.2 Case Studies Selection

When discussing the indicators of campus sustainability, it should be kept in mind that universities differ in terms of their physical characteristics and therefore have different constraints and drivers of implementation. For example, universities located in rural areas over hundreds of acres of land can afford to have organic gardens or renewable installations on their campus grounds that can reduce their carbon footprint. Those that are located in urbanized areas, and especially in big cities, do not have those capabilities and must focus on alternative measures for reducing their carbon footprint. In addition, campuses that have fewer students or are not research institutions implement different types of activities and have goals that are commensurate with their sustainability capabilities. Other differences stem from whether a university is public or private, whether it has a residential campus, and the type of degrees it offers.

Both AASHE and this chapter use the Carnegie Classification of Institutions of Higher Education to compare universities with similar characteristics, and to identify a sample of US universities (Indiana University Center for Post-Secondary

Research 2018). For this study, private, not-for-profit, Doctoral, four-year universities were selected as the main identifiers. The selection was narrowed to residential universities, given the consideration that universities providing housing to students living full time on campus affects the overall water and energy use, and consequently the type of sustainability efforts being pursued. To further narrow down this list, we selected only universities located in large cities of 250,000 residents or more because of the different types of challenges that rural and urban campuses experience, such as the size constraints of a city and the need to comply with city ordinances, to name a few. With these characteristics in mind, we identified 16 universities that were then cross-checked for valid sustainability ratings in the AASHE STARS classification. Nine of them had valid STARS scores, which were selected for analysis. They are presented in Table 1 and are discussed below.

The case studies methodology was identified as best suited for our research questions and type (Yin 2009). Specifically, case studies are most appropriate for examining “the exploratory and descriptive nature of research on the university-sustainability relationship,” for presenting current circumstances, and for explaining phenomena that require answers to many “how” or “why” questions (Karatzoglou 2013). The research questions here center on how sustainability offices and structures impact the effective implementation of campus sustainability and why some universities are more efficient in implementing sustainability than others. To investigate the differences among the types of offices, their organization, and the effectiveness of implementing campus sustainability, nine individual case studies were researched in detail.

Table 1 Selected universities for analysis and case studies. Data from the Carnegie Classification of Institutions of Higher Education and AASHE 2021 ratings

University	Location	Score	STARS rating	Valid until
American University	Washington, D.C.	72.66	Gold	April 2024
Carnegie Mellon University	Pittsburgh, Pennsylvania	69.69	Gold	February 2022
George Washington University	Washington, D.C.	68.2	Gold	March 2023
New York University	New York, New York	67.3	Gold	February 2023
University of Pennsylvania	Philadelphia, Pennsylvania	66.4	Gold	August 2021
Columbia University in the City of New York	New York, New York	65.89	Gold	August 2021
Case Western Reserve University	Cleveland, Ohio	58.96	Silver	August 2021
Boston University	Boston, Massachusetts	53.97	Silver	August 2022
Saint Louis University	Saint Louis, Missouri	34.71	Bronze	August 2021

6 Case Studies Presentation

6.1 American University, Washington, D.C.

In 2018, American University (AU), one of the leaders in HEI sustainability efforts in the United States, became the first urban campus of a research university to reach carbon neutrality. The Office of Sustainability, which has worked toward that goal since it opened in 2009, currently employs three full-time employees and student interns. All employees of the office are tasked with the implementation of the university sustainability plan. They report directly to the office of the Assistant Vice President of Planning and Project Management (American University 2021a). The Office of Sustainability works with AU's Facilities Management. Until recently, the overarching committee that devised and organized the sustainability plan and strategic direction of sustainability at AU was the Sustainability Project Team.

In April 2021, AU released a new sustainability plan that formalized this committee into a Sustainability Advisory Committee whose tasks are to review sustainability goals, actions, and issues. With the new sustainability plan, AU created the Center for Environment, Community, and Equity (CECE) which will serve as a central unit on campus focused specifically on the environment. Its mission statement is to “incubate innovative ideas, generate impactful and equitable actions and programming, and work to prepare a more diverse and inclusive next generation of thought leaders to advance a just transition to global sustainability” (American University 2021b). This center aims to create a space on campus where students can connect with AU's sustainability efforts and engage with different events and academic programs. By targeting a campus culture of sustainability through the Office of Sustainability and also via coordination with the department of facilities management, AU is able to successfully streamline sustainability efforts and meet the goals presented in their sustainability plan.

6.2 Carnegie Mellon University, Pittsburgh, Pennsylvania

Carnegie Mellon University (CMU), a STARS Gold Rating holder, has a strong focus on integrating sustainability into its campus culture and curriculum by meeting the UN SDGs and stressing the importance of sustainability in its mission statement and activities. Sustainability is a core value in the university Strategic Plan 2025, which was adopted in 2015 after holding more than 100 committee and subcommittee meetings (CMU 2021b).

CMU relies on two main committees for handling sustainability issues. One of them, the Environmental Practices Committee (EPC), convened in 1999 by the president of the university, aims to enhance the scope and impact of environmental education and research programs. The second one, the Green Practices Committee, is responsible for campus operations and for developing university practices that improve environmental quality, decrease waste, and conserve natural resources and energy. The university employs several facility-related staff, but only one

coordinator position with responsibilities focusing directly on sustainability and green practices. That position is located in the Facilities Management and Campus Services Department and is tasked with overseeing campus sustainability operations and engagement as well as coordinating and supporting the implementation of university green projects (AASHE 2019a). Thus, although CMU has a sustainability coordinator and several interns that work in the office, it does not have a designated sustainability office.

Regarding academics, CMU relies on the Steinbrenner Institute for Environmental Education and Research, launched in 2004, “to coordinate and advance the efforts at the intersection of research, education, and practice” (CMU 2021a). Consistent with the SDG emphasis, Carnegie Mellon’s engagement programs encourage collaboration with community partners to support social equity and well-being, economic prosperity, and ecological health.

6.3 George Washington University, Washington, D.C.

The George Washington University (GWU) is a STARS Gold Rating holder. Its location in the heart of Washington D.C. does not allow much room for green spaces, but it finds sustainability strength in academics, engagement, and sustainable building design. Its Office of Sustainability has six full-time staff members. Their responsibilities include determining the sustainability strategy and managing engagement with stakeholders. Two of the six staff members work with the university in an academic capacity, one serving as the Sustainability Minor program director and another as a research professor for sustainability initiatives (GWU 2020). Having played a major role in developing the strategic plan, they collaborate with a variety of campus offices to identify target initiatives, including collaborating with stakeholders in environmental education.

A variety of joint committees help to inform sustainability efforts at GWU, such as the Student Green Roundtable – where the Student Association and the Sustainable GWU office convene sustainability-focused student groups – and the Sustainable GWU Faculty Committee tasked with coordinating research, engagement, and academic initiatives. On the operations side, the SustainOps committee enables staff members across campus operation divisions to coordinate operational efforts to meet sustainability goals. The Committee for Sustainable Investment helps inform the university Sustainable Investment Fund. Lastly, the Climate Resilience Committee “provides input to and feedback on GWU’s climate resilience goals and targets” (AASHE 2020a). All of these committees comprise students, faculty, and staff. The Office of Sustainability is a member of these groups, enabling them to have greater influence and efficiency in sustainability programming at the university.

6.4 New York University, New York, New York

New York University (NYU) submitted its sustainability report to AASHE in 2020 and received a Gold rating with a score of 67.3. The Office of Sustainability, established in 2006, employs 10 staff members with the chief sustainability officer in the leadership role. The office, which reports directly to the president, is primarily responsible for campus sustainability operations and engagement (New York University 2020). Through collaboration with other stakeholders, it plays a major role in coordinating sustainability efforts in teaching and research.

NYU has several sustainability coordination groups. In September 2018, the university created a Sustainability Working Group convening students, faculty, and administrative staff to discuss Teaching & Research, Engagement, Energy, Water & Health, Food & Waste, and Transportation, and to identify new initiatives (AASHE 2020b). A Sustainability Advisory Group meets at least once per semester to enable administrators, students, and university senate representatives to assist in the general advancement of campus sustainability. Via the E-committee, student sustainability leaders are able to serve as peer sustainability mentors and act as liaisons to the Office of Sustainability, thus amplifying the student voice in campus sustainability work. The involvement of the Office of Sustainability in these groups enables stronger and better-informed sustainability efforts. With an emphasis on student voice, the office is able to create a campus environment where the culture of sustainability is better integrated throughout.

6.5 University of Pennsylvania, Philadelphia, Pennsylvania

The University of Pennsylvania (Penn) earned a score of 66.40 with its 2018 STARS report and was awarded a Gold rating. The Penn Sustainability Office has three full-time employees and three interns, who are all led by a sustainability director. Housed under the Facilities and Real Estate Services department, the office works to engage community members, manage campus operations, and collaborate with stakeholders via its membership in joint committees (Penn Sustainability 2020).

The main sustainability committee at Penn is the Environmental Sustainability Advisory Committee (ESAC), which is composed of students, faculty, and staff. Its role is to advise the president and senior leadership on sustainability issues with involvement from the Sustainability Office, operations teams, and academic centers. The recommendations from ESAC ultimately get integrated into the university's Climate Action Plan. It is also worth noting that beyond the coordination at the senior strategic decision-making level, Penn also hosts smaller committees to align campus realities with the strategic plan. The Student Sustainability Association as well as the Staff and Faculty Eco-Reps Steering Committees also help to bring about a culture of sustainability on campus. Additionally, sustainability plays a significant role in the decision-making process of all committees related to operations, such as the Bicycle Committee, the University Council's Facilities Committee, the Solid Waste Management Working Group, and the Tree Advisory Committee (AASHE 2018c).

Ultimately, through the large strategic development group ESAC, the Sustainability Office is able to collaborate with other university efforts, which leads it to participate in conversations around education and research.

6.6 Columbia University, New York City, NY

Rated Gold on the AASHE STARS classification, Columbia University is a leader in campus sustainability. Its plans integrate sustainability across campus academics, operations, and engagement into one organizational structure. In 2016, three focus teams were created to “centralize the sustainability conversation between academic and administrative leaders on campus” (Columbia University 2021b). These three focus teams brought together students, faculty, and administrators to direct the overall strategic sustainability plan and future mission of Columbia University. The Chair(s) of each of these teams sits on the Senior Sustainability Advisory Committee (SSAC) and with other team members oversees and tracks progress on the campus climate goals. The SSAC then looks to the Office of Environmental Stewardship, the Earth Institute, and programs such as the Sustainable Leaders Network, to implement the sustainability goals.

Columbia’s sustainability-focused academics and research are overseen by the Earth Institute, which administers 24 environmental degrees. These degrees are housed in multiple University schools, but are closely affiliated with the Earth Institute to ensure a high degree of coordination and collaboration within sustainability-focused academic programs (Columbia University 2021a). This coordination is unique in that the degrees themselves are not housed within the Earth Institute but are spread throughout the schools.

6.7 Case Western Reserve University, Cleveland, Ohio

Case Western Reserve University (CWRU) places emphasis on imbuing sustainability in its mission statement, overall strategic plan, and its campus. The Office of Energy & Sustainability has two full-time employees and is separate from facilities and academics, but it closely coordinates with them to ensure the implementation of the University’s Climate Action Plan and the American Colleges & Universities Presidents’ Climate Commitment (ACUPCC). In 2020, CWRU began building a Climate Action Network to unify the university’s sustainability efforts by connecting sustainability-related activities across campus and increasing the coordination of different climate actions.

There is a strong desire for building a culture of sustainability, which is visible in the university academic and engagement programs, and movement toward a more integrated sustainability structure. The Sustainability Alliance, which was created in 2009, coordinates academic sustainability by working in three main areas: research and scholarship, education, and community connection (AASHE 2018a). By serving as a coordinating force, this alliance further complements the goals of encouraging

sustainable practices as a way of life, consistent with the university as a Living Lab model.

There are multiple opportunities for students, faculty, and staff to engage with sustainability. For example, the Climate Action Plan is emphasized as a “living document,” one that requires innovation and evolution as “ideas and technology change” (Case Western Reserve University 2011, p. 22). The movement toward a more integrated structure and focus on community involvement encourages students and faculty to feel empowered to participate in the overall community sustainability commitment and allows for continual growth and refinement of the university policies.

6.8 Boston University, Boston, Massachusetts

The sustainability office of Boston University (BU), called BU Sustainability, is housed under Facilities Management & Planning and reports to the Senior Vice President of Operations. BU Sustainability has seven full-time staff members. Their responsibilities cover areas such as communications, data management, engagement, zero waste, and dining services. The office also employs 10 student interns. Although not directly responsible for academic programming on campus, BU Sustainability helps inform academics on matters related to sustainability through the sustainability task force. It is also a part of the Climate Action Plan Task Force, which was convened by the president to develop the campus action plan that has three goals: minimizing emissions, addressing climate resilience, and enhancing education and research activities (AASHE 2019b).

Boston University has several institutes and centers that support sustainability in education and research. BU engages students in sustainability through different means, such as student clubs, hiring interns in the sustainability office, and the Earth House Living Learning community, “which integrates curriculum with student life and housing” (Earth House Living Learning Community). To further extend sustainability to student housing, the university has also launched a Resident Sustainability Leaders program to work alongside Resident Assistants (RAs) and Residence Hall Council (RHC) to promote sustainability in student living in general. BU also has a green office program that engages faculty and staff.

6.9 Saint Louis University, Saint Louis, Missouri

Saint Louis University (SLU) received a Bronze rating from AASHE for its 2018 sustainability report. The Office of Sustainability & Benchmarking, which is housed under Facilities Services, is in charge of identifying, prioritizing, coordinating, and tracking campus sustainability efforts. It has two full-time staff members – a project analyst and a sustainability coordinator. SLU utilizes an external consult, Sightlines Consulting Group, to help evaluate its sustainability performance. Due to the small team of two people and the scope of work of the office, it does not have specific

responsibilities in shaping the curriculum (AASHE 2018b). Through the Sustainability Advisory Council, a joint effort of the Vice President for Facilities Services and the Director of the Center for Sustainability, stakeholders across the university are able to coordinate sustainability efforts and ensure sustainability integration into academic, administrative, and operational practices (AASHE 2018b). Ultimately, this allows for a higher degree of efficiency in campus sustainability.

7 Findings from the Case Studies and Discussion

The case studies presented here constitute an overview of sustainability office structures and practices and show that, on average, those under structure Type C exhibit higher STARS ratings than those under structure Type B, which in turn have higher STARS ratings than those in structure Type A. Based on the definitions for the three structure types and the information obtained about the nine universities' office and organizational structures, we categorized the universities and their sustainability office structures in the following way: Saint Louis University as structure Type A; Carnegie Mellon University as trending from structure Type A to B; Case Western Reserve University and Boston University as structure Type B; American University, George Washington University, New York University, and University of Pennsylvania as trending from structure Type B to C; and Columbia University as structure Type C.

Under all structure types, universities work with and involve stakeholders to advance sustainability efforts. The differences stem from the extent and type of engagement. While universities under structure Types A and B typically have committees with varying degree of coordination between their activities and efforts, their functions and organizations remain relatively compartmentalized. For example, the office of sustainability at CWRU, which falls under a typical Type B structure, has several committees; moreover, Case Western has various research centers devoted to studying sustainability solutions, such as the Great Lakes Energy Institute and the Mary Ann Swetland Center for Environmental Health. However, it was not until 2020 that CWRU began building a Climate Action Network to unify the university's sustainability efforts by connecting sustainability-related activities across campus and increasing the coordination of different climate actions. Creating new, more unifying structures that communicate, coordinate, and serve to better represent the many different centers, institutes, committees, and stakeholders underlines the evolving nature of university office structures and governance; such developments also reinforce what many papers state: that the case studies, despite their many advantages for detailed representations of reality, constitute a snapshot in time (Karatzoglou 2013; Lozano et al. 2015). Despite this limitation, case studies are useful for creating benchmarks that enable institutions to track their progress, identify and document successes, and manage the levels of resources devoted to (and required for) the attainment of sustainability goals.

In that sense, credit is due to organizations like AASHE that have developed a mechanism to collect and evaluate data, and that have created a repository of

systematically gathered data that can be used both for tracking progress and for comparative analysis. The fact that the literature has not settled on a unified set of campus indicators for organizing sustainability structures and practices shows that campus sustainability is still a work in progress. To underscore the transitional nature of campus sustainability, it needs to be pointed out that the STARS scoring system and even the reporting for the scoring categories have been evolving and becoming more comprehensive. Since the initial idea for developing a campus rating system in 2006, and the first release of the STARS technical draft manual in 2007, the manuals have undergone five revisions to the current 2.2 version, which was released in June 2019 (AASHE 2019).

Increased communication and coordination allow for more transparency and synergy across the campus and lead to more effective results, which is the driving impetus for many Type A universities to be trending toward a Type B structure, and for Type B to be trending to Type C. The latter trend characterizes universities such as AU, NYU, Penn, and GWU. All four universities have Climate Action Plans and are working inclusively via sustainability committees to advise the Presidents of their respective universities on sustainability issues. Of these four universities, the one that is a bit of an outlier is AU. American University has the highest STARS score of the examined sample of universities; despite this rating, it was not categorized as a Type C structure. Instead, it was characterized here as trending toward a Type C structure because, when compared to other Type C universities, AU tends to have a less cohesive sustainability organization – for example, AU has separate departments in charge of different aspects of sustainability. However, with the creation of its new Center for Environment, Community, and Equity, which was recently announced (April 2021), AU aims to integrate its sustainability plan in terms of academics, research, and engagement in order to foster a sustainability campus culture and, therefore, will soon be categorized as a Type C structure.

In addition, the movement from compartmentalization to consolidation follows trends already described in the literature. In their historical overview of campus sustainability, which identifies three distinct periods, Washington-Ottombre et al. (2018) state that sustainability efforts of HEI started to move away from being compartmentalized and implemented as “add-on” initiatives only after 2010, around which time they started becoming more integrated and more holistically viewed as a “concerted effort geared toward impacting every facet of HEIs” (p. 571). Such trends are necessitated by stakeholder calls for integrating sustainability education into policies and practices, on one hand, and by efforts on the part of the universities themselves for institutionalizing sustainability, on the other hand.

7.1 Limitations

These changes suggest that some universities are recognizing the need to more closely coordinate sustainability efforts, which can most efficiently be done if sustainability efforts are overseen by a single committee or office. Having a functionally integrated office structure is recommended because issues can be more

quickly addressed and decisions can be streamlined (Lozano et al. 2015). A question that logically follows from this is: Are universities that have a diffuse sustainability structure and that do not have a central office as effective in implementing sustainability as those that have a centralized structure? This chapter is not suited for answering this question, as it has several limitations. First of all, nine universities constitute a very small sample from which to draw overarching conclusions. Second, the findings discussed here need to be validated via questionnaires or surveys (for example, of a university's stakeholders) before any conclusions could be drawn about the overall impact of this research. The third limitation stems from the nature of the methodological approach selected for analyzing the research question, namely using case studies.

7.2 Further Research

Further research is needed to investigate the remaining questions that this research poses. For example, it will be interesting to find out whether the three proposed structure types can be used as indicators to show which universities will tend toward what structure type in their next reports. For that type of analysis, it will be useful to review universities that have a Platinum STARS rating and to include a larger number of those with Bronze and Silver ratings to see how they differ from those that have Gold ratings. It was only 6 years ago that Colorado State University became the first institution to earn a STARS Platinum rating. Others have followed suit and especially in the last few years, universities are changing at a faster rate than they have been before (AASHE 2021). In that respect, this chapter joins others like Leal Filho et al. (2020) who call for longitudinal studies, which could be useful to understand how several variables change over time, thus helping us understand the evolution of sustainable organizational structures and leadership. Such longitudinal analyses could shed light on whether universities with higher STARS ratings have been participating in AASHE and monitoring their sustainability efforts for a longer time. For example, the AASHE Sustainability Tracking, Assessment, and Rating System shows that the universities that are Platinum-holders have submitted between two and six reports before obtaining the Platinum rating. If that is correct and if efforts toward consolidation and coordination continue, would the sustainability structure types proposed in this chapter be correct in pointing to trends about who the next Platinum holders will be? More research is needed to validate such trends and develop frameworks that could establish best practices.

8 Conclusion

This chapter has presented nine case studies from universities with different organizational structures. It explores the relationship between office structures and performance, measured by the universities' STARS ratings. The proposed framework can now serve as a first step toward analyzing university organizational structures and

discussing best practices in their functions. There is a need for further research and for triangulation of the results, which calls for conducting surveys and interviews with both the people who are in charge of creating those structures and with the participants in them.

Conceptualizing this chapter has been challenging as there are many gaps and many uncertainties about what should constitute best practices of office structures. Some gaps stem from the lack of an overarching vision about integrating operations with academics to more holistically look at where campuses need to be in 20–30 years from now. The UN SDGs were developed in 2015 and called for a 15-year horizon for achievement. What would happen afterward? What goals should be set by universities that can help implement the transition to sustainability? How should campus operations and leadership look in 2050?

Creating indicators for measuring sustainability is not an easy task. Designing metrics and practices should be done by anticipating what is most relevant to the future of universities, to their stakeholders, and to future generations. Perhaps, performance indices should include a measure for the number of jobs graduates are able to get in sustainability or in promoting sustainable practices.

Finally, it needs to be recognized that campus sustainability structures do not operate in a vacuum; they depend on and interact with many other elements that constitute a sustainable campus. Although we tried to select universities that have similar physical characteristics for this chapter, there are many other factors that impact sustainability implementation at universities as well. Examining the role of leadership, funding, and networking, to name a few examples that have already been discussed in the literature, can assist to improve sustainability performance of higher education institutions.

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